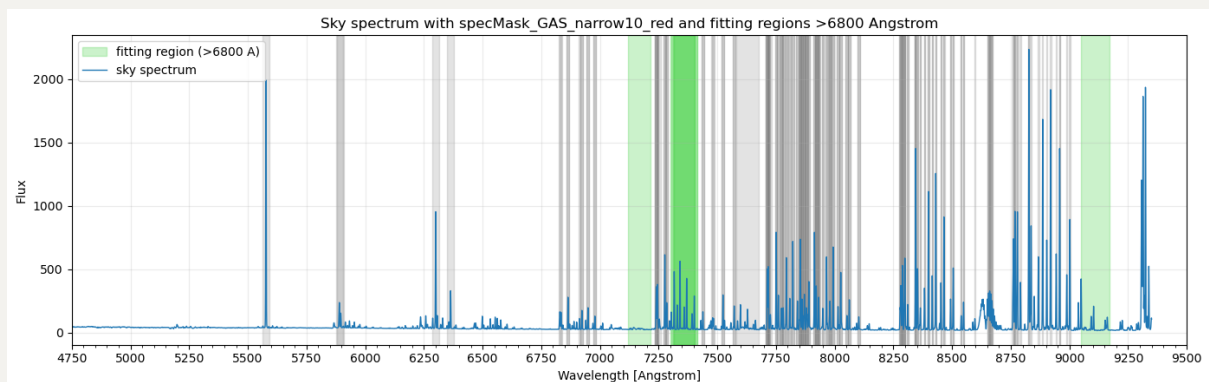
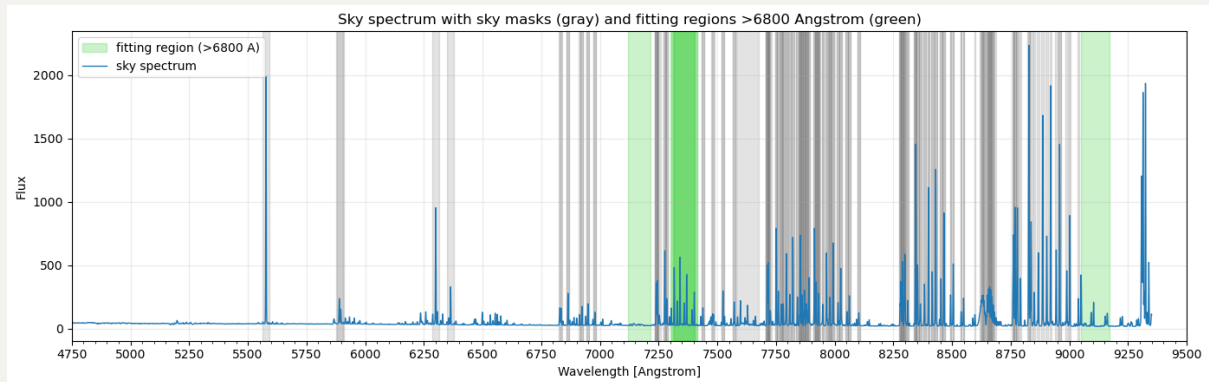


20260517 Original spectral masking for spectra higher than 8250Å

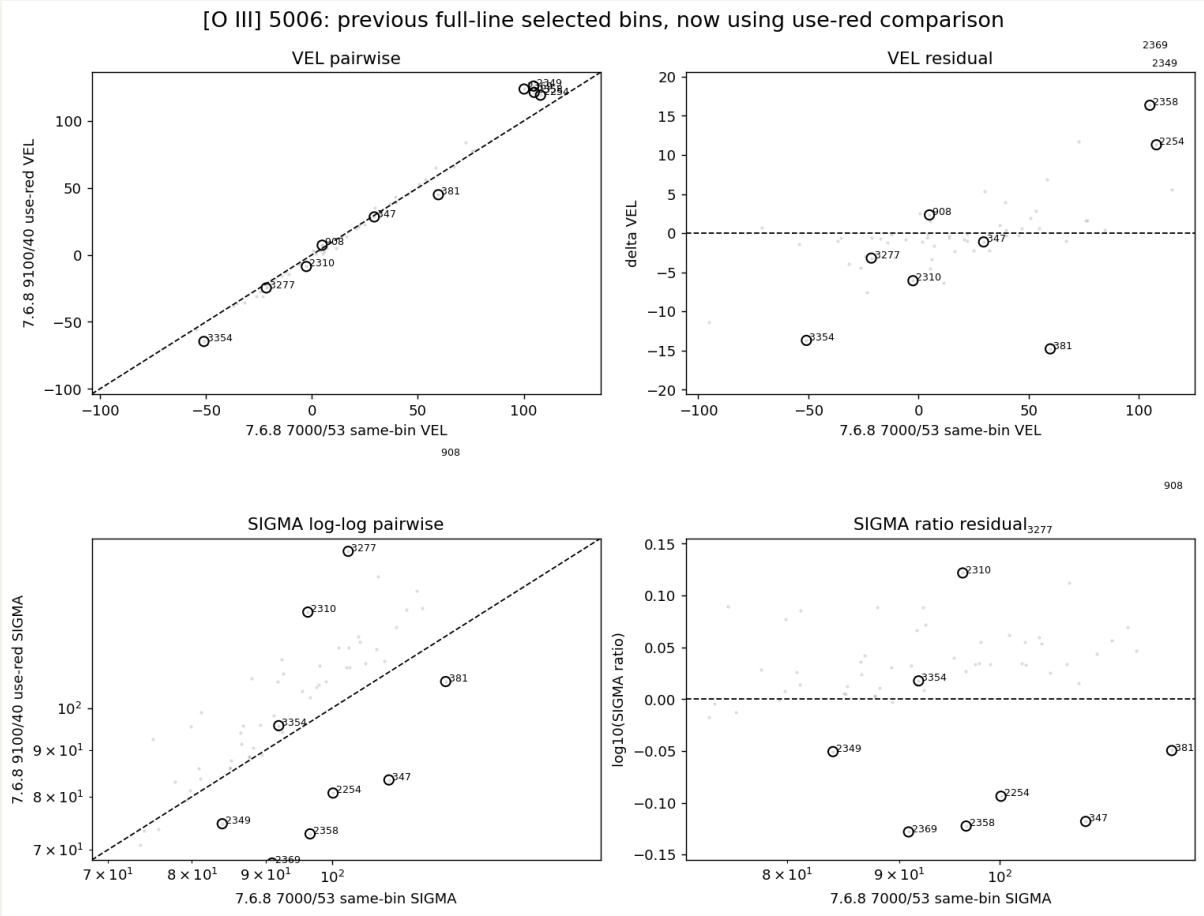
Here we test if the 9100/40 rerun keeps the used-line gas list, including SIII9068, but relaxes the red-end ($> 8250\text{\AA}$) spectral masking via specMask_GAS_narrow10_red, the OIII5006 high-S/N outlier behavior should show whether the relative velocity and sigma residuals persist against the 7000/53 same-bin reference.

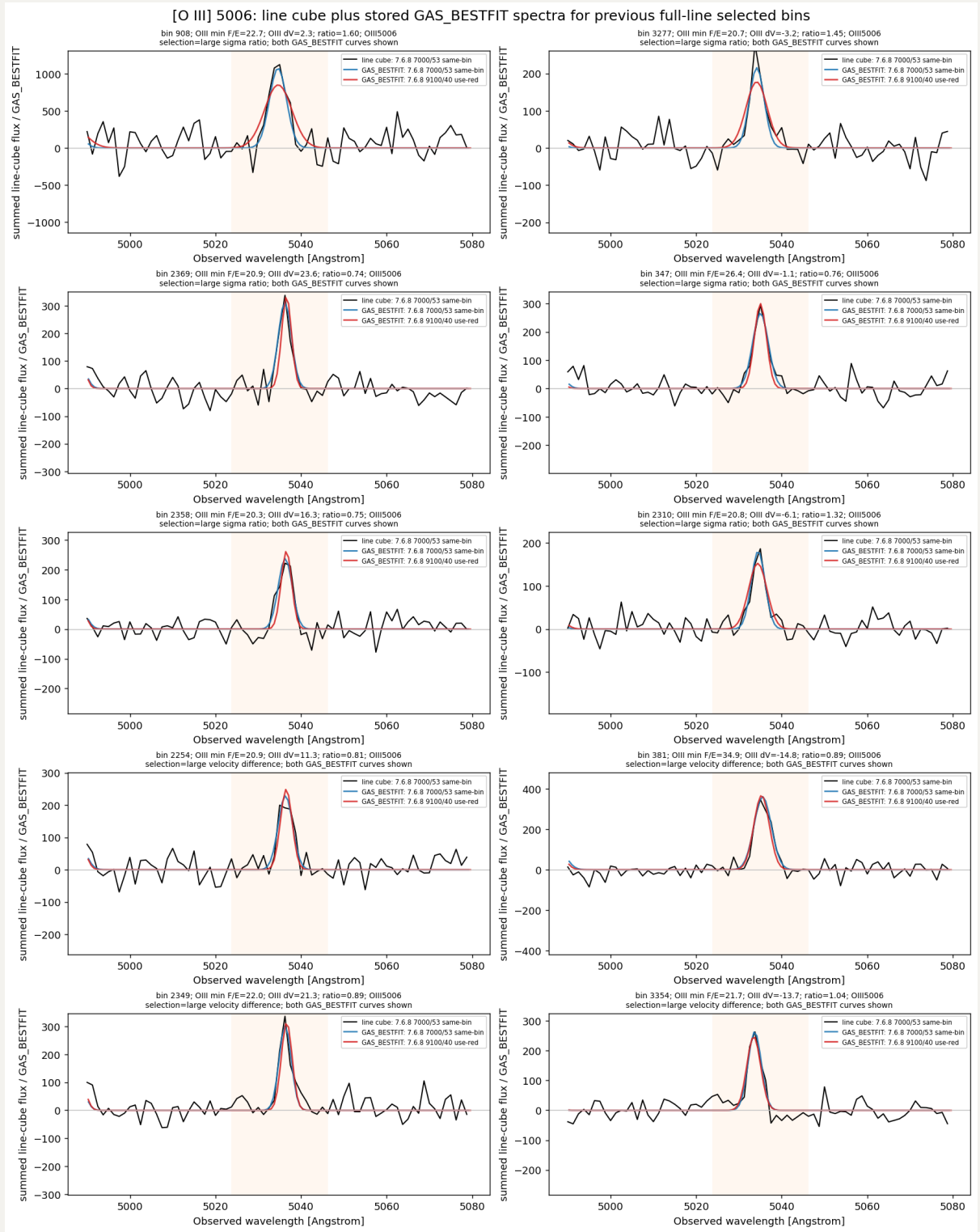
First, we show the spectral masking of previous specMask_GAS_narrow10 and specMask_GAS_narrow10_red. The updates are:

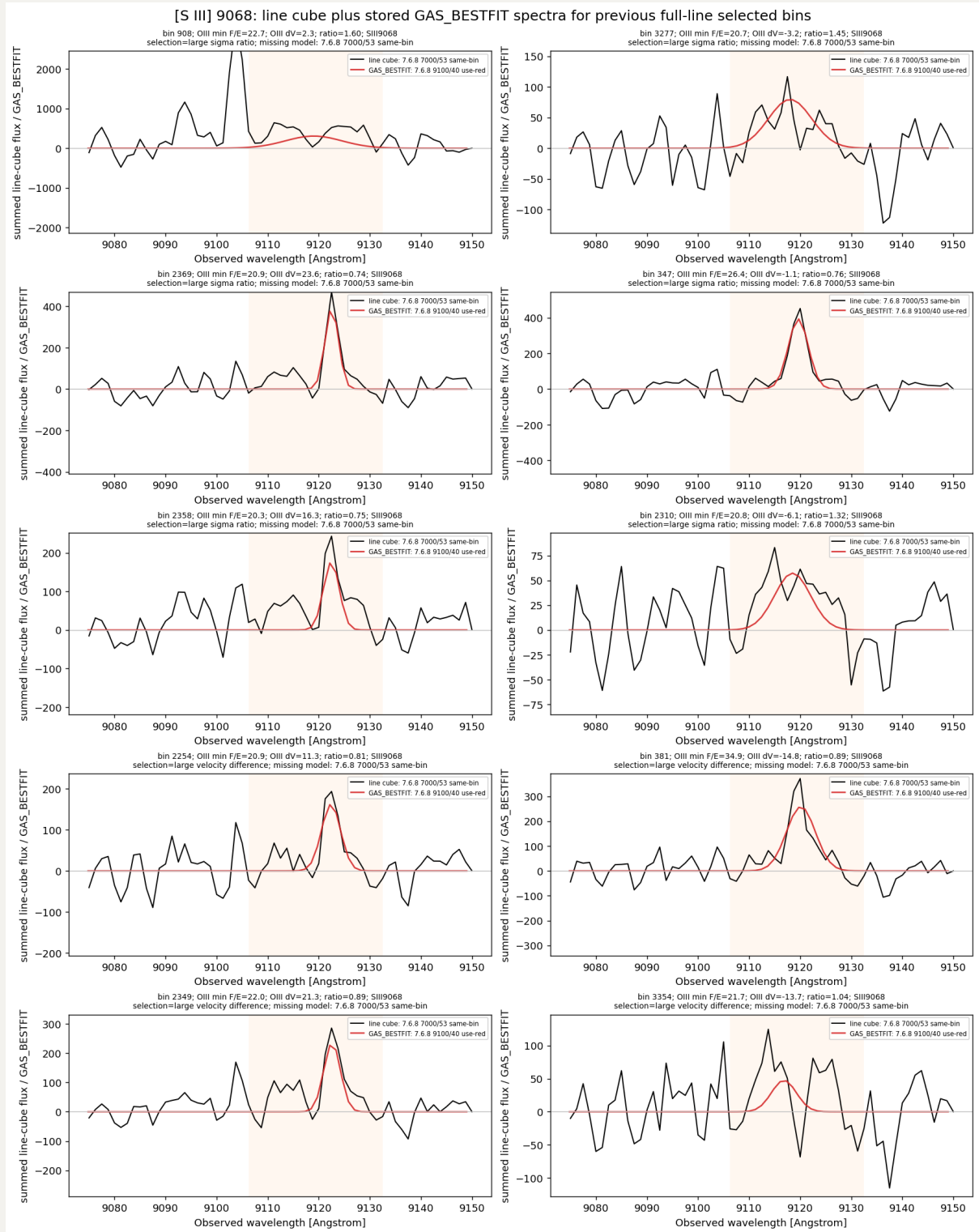
- $6800 < \lambda \leq 8250\text{\AA}$: still uses specMask_KIN_narrow10.
- $\lambda > 8250\text{\AA}$: uses the original specMask_KIN red sky masks, preserving 5 Angstrom widths.
- No extra detected-peak masks from specMask_KIN_narrow10 are added above $8250\text{\AA}$.
- Protected emission line fitting regions (green) are still left unmasked.



Then, we check those 10 selected bins. Unfortunately, this method still cannot fix the biased [O III] λ 5007 kinematics issue.







So we should think about the potential way to really fix this.

One is just easily modifying the `ngist`'s wrapper to make it adapt more gas fitting components so that we can fit those faint but useful lines in `free` mode and so does not affect `OIII5006`'s kinematics.

The other is that we still input 9100 spectra but only fit up to $[\text{O II}]\lambda\lambda 7320, 30$, and then for the regions that `SIII6312` is detected, we do an extra `ppxf` fitting for it to include `SIII9068`. For this approach, we need to be careful with two kinematics outcomes of `SIII6312`: one ties to `OIII5006` and the other ties to `SIII9068`. I think we may fix the kinematics of the extra fitting as the outcome of previous one, but doing so may also require modification on `ngist`.